



Bridging Project 1 — RLHF / DPO on Small Base Models

Goal

Build, train, and ship a complete RLHF/DPO pipeline that takes a small base model from SFT through preference optimization, with rigorous comparison between PPO and DPO, on a real preference dataset. Document the experimental methodology, results, and honest failure modes in a research-quality writeup.

This is one of three bridging projects designed to close the gap between current portfolio and target lanes (post-training engineer at foundation labs, Applied AI at fine-tuning service companies).

Why this project — the career signal

What it signals to hiring managers:

- I can take an open preference dataset, build a full RLHF pipeline (reward model + policy training), and ship working artifacts in days, not weeks.
- I understand post-training methodology end-to-end: SFT baseline → reward modeling → DPO → PPO → evaluation.
- I can compare approaches rigorously and report findings honestly, including what failed.

What it opens up:

- Post-training engineer roles at Anthropic, OpenAI, Mistral, Cohere, AI21
- Applied AI roles at OpenPipe, Predibase, Together AI (fine-tuning services)
- MTS roles at frontier labs where preference optimization is part of the job

Scope

One repo, four trained artifacts, one Gradio demo, one writeup.

In:

- SFT baseline on a small base model (Qwen2.5-1.5B or Llama-3.2-3B)
- Reward model trained on a public preference dataset (UltraFeedback recommended)
- DPO training using the same preference data, LoRA adapters
- PPO training using the reward model — the harder leg
- Side-by-side evaluation on AlpacaEval 2.0 and MT-Bench
- Gradio demo comparing the three outputs on the same prompt
- Writeup with results table and honest failure-mode discussion

Out of scope:

- Constitutional AI / RLAI (additive complexity for no signal gain)
- Distributed training (single-node is fine)
- Custom reward model architectures (use existing patterns)
- Online RLHF (offline only)

Dataset choice

Pick one:

Dataset	Size	Notes
Anthropic HH-RLHF	~160k pairs	Classical, easy comparison vs. published baselines
UltraFeedback	~64k pairs	GPT-4 annotated, higher quality, used by Zephyr
HelpSteer2	~10k samples	Multi-attribute ratings, needs conversion to pairwise

Recommended: UltraFeedback for quality/iteration speed tradeoff.

Base model

Model	Why	When
Qwen2.5-1.5B	Fast iteration on A10G	Recommended primary
Llama-3.2-3B	Strong open base, well-supported by <code>trl</code>	Optional second run for cross-model story
Qwen2.5-3B	Newer, very capable	Skip unless time permits

Training stack

- `transformers` ≥ 4.45
- `trl` ≥ 0.11 (DPOTrainer, PPOTrainer, RewardTrainer)
- `peft` for LoRA adapters
- `accelerate` for multi-GPU
- `bitsandbytes` for 4-bit if memory-constrained
- `wandb` for experiment tracking

Evaluation methodology

- **AlpacaEval 2.0 (length-controlled win rate vs SFT baseline)** — the standard metric
- **MT-Bench (GPT-4 judge)** — multi-turn capability check
- **Reward model accuracy on held-out preferences** — sanity check on the RM itself
- **Reward + KL curves during PPO** — diagnostic for reward hacking

Cost & infrastructure

All on RunPod community cloud unless H100 noted:

Phase	Hardware	Time	Cost
SFT baseline	A40 48GB	30-60 min	~\$0.40
Reward model training	A40	1-2 hr	~\$1.00
DPO training	A40	2-4 hr	~\$2.00

Phase	Hardware	Time	Cost
PPO training	A100 80GB	4-6 hr	~\$10.00
Eval runs	A10G	1-2 hr	~\$0.80
Iteration buffer (2x)	—	—	+100%

Total realistic budget: \$30-50 on RunPod, plus \$20-40 in GPT-4 judge API costs for evals.

Timeline — 5 days at demonstrated cadence

Day	Work
1	Repo scaffold. Data prep (UltraFeedback → pairs). SFT baseline run.
2	Reward model training. Sanity-check held-out accuracy.
3	DPO training. First evaluation pass on AlpacaEval.
4	PPO training (the hard one). Debug reward hacking if it appears.
5	Final evals. Gradio demo. Writeup. Push everything to HF.

Definition of done

- ☐ Repo on GitHub with clean module structure (`src/sft/` , `src/rm/` , `src/dpo/` , `src/ppo/` , `src/eval/`)
- ☐ Three LoRA adapters pushed to HuggingFace Hub (SFT, DPO, PPO)
- ☐ Reward model pushed to HuggingFace Hub
- ☐ Gradio Space deployed with side-by-side comparison of all three outputs
- ☐ README with results table, training cost, key findings, honest discussion of what didn't work
- ☐ At least one non-obvious learning documented (e.g., why DPO hyperparameters didn't transfer cleanly)
- ☐ No "WIP" components — finish or remove

Risks & failure modes

- **PPO instability.** Reward hacking, mode collapse, KL explosion are real. Plan: start with conservative KL coefficient, monitor reward + KL separately. If PPO fails after 2 attempts, document the failure honestly in the writeup — that is itself a strong signal.
- **Reward model overfitting.** Small RMs trained on small data can hit 95% train accuracy and 65% test. Use a held-out set; report both numbers.
- **AlpacaEval and MT-Bench cost.** Both require GPT-4 as judge. Budget \$20-40 for API calls.
- **DPO plateau.** If DPO loss plateaus quickly, the preference data is too easy. Switch to a harder split or different dataset.

Reading list

Required before starting:

- DPO paper (Rafailov et al., 2023)
- InstructGPT paper (Ouyang et al., 2022) for the canonical RLHF pipeline
- HuggingFace `trl` documentation for DPOTrainer and PPOTrainer
- The Zephyr training writeup (HuggingFace blog)

Nice to have:

- Constitutional AI paper (Bai et al., 2022)
- UltraFeedback paper (Cui et al., 2023)
- Anthropic HH-RLHF dataset card

What this should NOT be

- Not a "ran the tutorial" project. Tutorials cover SFT + DPO. The differentiator is the PPO leg, the honest comparison, and the failure-mode documentation.
- Not a comprehensive survey. Pick one dataset, one base model. Depth over breadth.
- Not a research contribution. The goal is to demonstrate competence, not novelty.